

Practice Midterm 1

Math 124, Fall 2026

Name: _____

username: _____

UMID: _____

Room: ☐ B844 East Hall

Instructions

- This exam consists of 7 problems across 10 pages, worth a total of 100 points.
- You have 120 minutes to complete this exam, unless you have extended-time accommodations through SSD.
- Write your username in the top right corner of each page where a space is provided.
- For free response problems, **you must show all of your work**, and write your final answer in the small box when one is provided.
- For multiple choice problems, completely fill in bubbles and square boxes.
 - ☐ A bubble means that you should only select one choice.
 - ☐ A square box means that you should select all that apply.
- You may refer to **one double-sided 4×6 -inch index card**, handwritten directly on the card. Other than that, you may not refer to any other resources or technology during the exam (no phones, watches, or calculators).

By signing below, I affirm that this work is my own and that I have followed the exam's resource and collaboration rules.

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Problem 1 (9 pts)

Consider the points

$$P = (-7, 8, -6), \quad Q = (8, -12, 19)$$

- a) (2 pts) Let $\vec{d}$ be the vector pointing from P to Q . Find $\vec{d}$. Give your answer as a vector.

$\vec{d} =$

- b) (3 pts) Find the distance between P and Q .

distance =

- c) (4 pts) Find the coordinates of the point R that is $\frac{3}{5}$ of the way from P to Q .

$R =$

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Problem 2 (16 pts)

Suppose $\vec{u}, \vec{v} \in \mathbb{R}^3$ satisfy

$$\|\vec{u}\| = 5, \quad \|\vec{v}\| = 4, \quad \vec{u} \cdot \vec{v} = -6$$

a) (7 pts)

I) (5 pts) Find $(5\vec{u} - 6\vec{v}) \cdot (8\vec{u} + 10\vec{v})$. Show your work.

dot product =

II) (2 pts) What type of angle is formed by $5\vec{u} - 6\vec{v}$ and $8\vec{u} + 10\vec{v}$? Select one.

☐ Acute ☐ Right ☐ Obtuse

b) (5 pts) Find $\|\vec{u} - \vec{v}\|$.

$\|\vec{u} - \vec{v}\| =$

c) (4 pts) Could a unit vector $\vec{w} \in \mathbb{R}^3$ satisfy $\vec{u} \cdot \vec{w} = 9$? Explain why or why not.

Problem 4 (14 pts)

The vectors

$$\vec{u}_1 = \frac{1}{19} \begin{bmatrix} -15 \\ 10 \\ 6 \end{bmatrix}, \quad \vec{u}_2 = \frac{1}{19} \begin{bmatrix} -6 \\ -15 \\ 10 \end{bmatrix}, \quad \vec{u}_3 = \frac{1}{19} \begin{bmatrix} 10 \\ 6 \\ 15 \end{bmatrix}$$

form an orthonormal basis of $\mathbb{R}^3$.

- a) (2 pts) Given that these vectors form an orthonormal basis of $\mathbb{R}^3$, find the following quantities.

$$\vec{u}_1 \cdot \vec{u}_3 = \boxed{} \quad \|\vec{u}_1\| = \boxed{}$$

- b) (4 pts) Find $\|5\vec{u}_1 - 7\vec{u}_2\|$.

$$\|5\vec{u}_1 - 7\vec{u}_2\| = \boxed{}$$

- c) (8 pts) Let $\vec{w} = \begin{bmatrix} -4 \\ 3 \\ 4 \end{bmatrix}$. Write $\vec{w}$ as a linear combination of $\vec{u}_1$, $\vec{u}_2$, and $\vec{u}_3$.

$$\vec{w} = \boxed{} \vec{u}_1 + \boxed{} \vec{u}_2 + \boxed{} \vec{u}_3$$

Problem 5 (19 pts)

Let P be the plane through the points

$$A = (8, -15, 6), \quad B = (13, -5, 11), \quad C = (15, -8, 7)$$

- a) (5 pts) Express P in vector-parametric form.

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} =$

- b) (8 pts) Write an equation for P in the form $ax + by + cz = d$.

equation:

- c) (6 pts) Suppose C is replaced by $C' = (23, 15, 21)$, while A and B stay the same. Do the three points A, B, C' determine a unique plane? Explain what changes geometrically.

Problem 6 (18 pts)

Consider the line ℓ in $\mathbb{R}^3$ given by

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ -8 \\ 11 \end{bmatrix} + t \begin{bmatrix} 5 \\ -6 \\ 7 \end{bmatrix}, \quad t \in \mathbb{R}$$

- a) (10 pts) Find equations for two distinct planes whose intersection is exactly ℓ . Your equations should both be of the form $ax + by + cz = d$.

plane 1:

plane 2:

- b) (8 pts) A student proposes the equations $11x + 15y + 5z = 12$ and $55x + 75y + 25z = 60$. Every point on ℓ satisfies both. Explain why these equations, unlike those in part (a), do not intersect exactly at the line ℓ . Describe their intersection and give one point in it that is not on ℓ .

one such point =

Problem 7 (10 pts)

Let

$$\vec{v} = \begin{bmatrix} 6 \\ 8 \\ -8 \end{bmatrix}, \quad \ell = \text{span} \left(\begin{bmatrix} 5 \\ -5 \\ 7 \end{bmatrix} \right), \quad P : 5x - 5y + 7z = 0$$

- a) (6 pts) Find the orthogonal projection of $\vec{v}$ onto ℓ .

$\text{proj}_{\ell}(\vec{v}) =$

- b) (4 pts) Find the orthogonal projection of $\vec{v}$ onto P .

$\text{proj}_P(\vec{v}) =$

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Congrats on finishing Practice Midterm 1!

Feel free to draw us a picture about Math 124 in the box below (or use it for scratch work).

A large, empty rectangular box with a thin black border, intended for a drawing or scratch work related to Math 124.